# Project Definition Document

**Course:** BIL 482 Design Patterns – TOBB University of Economics and Technology, Fall 2026

**Instructor:** Dr. Çiğdem Avcı

**Assignment:** Project Assignment 1 (PA1)

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## 1. Project Name

**Skill/Interest-Based Secure Matching Platform for Erasmus and Local Students**

## 2. Project Summary

Exchange programs aim to foster cross-cultural integration, but international students often end up socially isolated. Language and cultural barriers push Erasmus students into closed groups of other foreigners, and universities rarely offer a simple way to connect them with local students who share their interests.

This project is a multilingual web platform (Turkish, English, Spanish) that matches Erasmus and local students based on shared skills, hobbies and language goals rather than nationality. Users create a profile, find compatible students, create and join events, receive notifications, and report inappropriate content. The system is built on a Layered Architecture and applies design patterns (Factory Method, Strategy, Observer, Adapter and Chain of Responsibility) to keep it flexible, modular and easy to maintain.

## 3. Objectives

By the end of the project, the platform will:

1. Allow students to register and create profiles with their skills, hobbies and language proficiencies (UC1).
2. Match Erasmus and local students using configurable criteria such as language exchange, hobbies or sports (UC2).
3. Allow users to create and join events, and notify them about new matches and event updates (UC3).
4. Provide a reporting system for inappropriate content, escalated from automatic filtering to human moderation (UC4).
5. Offer the interface in Turkish, English and Spanish.
6. Apply at least one design pattern per use case, documented and traceable from design to implementation.

## 4. Scope

**Included:**

- User registration and profile creation (skills, hobbies, languages, matching preferences)
- Interest and skill-based matching with user-controlled filters
- Event creation, discovery and participation
- Notifications for new matches and event updates
- Reporting and moderation of inappropriate content
- Web interface in Turkish, English and Spanish
- eduroam authentication and translation service, accessed through Adapters (simulated with mock implementations)

**Not included:**

- Native mobile applications (iOS / Android)
- Real production integration with eduroam and a commercial translation API
- Real-time chat between users
- Payments or paid features
- Full legal GDPR/KVKK certification (considered only as design requirements)

## 5. Target Audience

- **Erasmus and exchange students:** incoming international students who want to integrate into the local community, overcome language and cultural barriers, and meet students with similar interests.
- **Local students:** students of the host university who want to practice foreign languages, take part in cultural exchange, or find partners for hobbies, sports and other activities.
- **Moderators / administrators:** university staff or student volunteers who review reported content and keep the platform safe.

## 6. Key Features

| # | Feature | Description | Related Use Case |
|---|---|---|---|
| F1 | User registration & profile creation | Register through eduroam or an invitation code and create a profile with skills, hobbies and languages | UC1 |
| F2 | Interest/skill-based matching | Find compatible students using a selectable matching criterion and personal filters | UC2 |
| F3 | Event management | Create, discover and join events and activities | UC3 |
| F4 | Notifications | Receive alerts about new matches and changes in joined events | UC3 |
| F5 | Content reporting & moderation | Report inappropriate content, handled by automatic filter, moderator and admin | UC4 |
| F6 | Multi-language interface | Interface in Turkish, English and Spanish, with translation of user content | All |

## 7. Deliverables

| Deliverable | Content |
|---|---|
| PA1 | Project Definition, Project Plan, Software Requirements Specification (with Software Architecture Overview and Pattern–Use Case Mapping) |
| PA2 | One selected use case with detailed pattern application, presented in the Design Pattern |

| Deliverable | Content |
|---|---|
| | Workshop |
| PA3 | Final project design and technologies |
| Source code | Implementation of UC1–UC4 with the documented patterns, in the GitHub repository |
| Final presentation & demo | Working demo of the platform and final documentation |

## 8. Project Success Criteria

1. All four core use cases (UC1–UC4) are fully implemented and work end to end in the final demo.
2. Every use case applies at least one design pattern, and each pattern can be traced from the documentation to the source code.
3. At least 80% of test users (Erasmus and local students) rate the platform as easy to use.
4. The interface is fully available in Turkish, English and Spanish.
5. A user can complete registration and receive a first match in less than 5 minutes.

## 9. Task Matrix

| Task | Responsible |
|---|---|
| Project Name and Summary | Alberto Alvarez Polo |
| Objectives and Scope | Alberto Alvarez Polo |
| Target Audience and Key Features | Alberto Alvarez Polo |
| Deliverables and Success Criteria | Alberto Alvarez Polo |
| Review of objectives and scope | Alejandro Blas Piris |
| Review of key features and success criteria | Carlos Molina Perez |
| Final formatting and integration of feedback | Alberto Alvarez Polo |